Overview

Your team will model, analyse and control an acrobot: a two-link pendulum with a motor at the elbow and none at the shoulder. The project runs in four phases. Phase 1 is graded and returned right away, so you can build the later phases on a correct model.

Tools: MATLAB.

The system

q1 is measured from straight down; q2 is the elbow angle relative to link 1. Only the elbow is actuated, by torque τ.

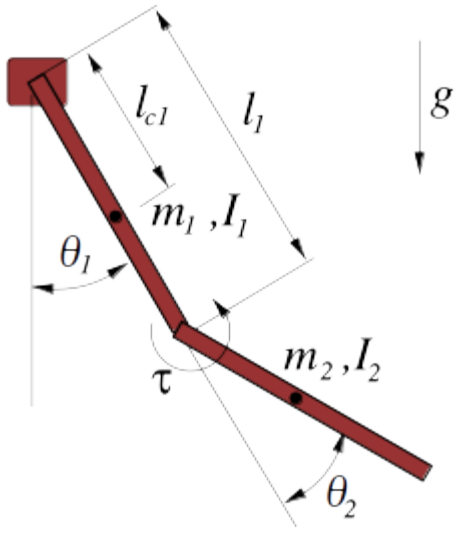


Figure 1: Taken from Underactuated Robotics, © Russ Tedrake, 2024

Parameters

Symbol	Meaning	Value
m_1, m_2	link masses	1 kg each
l_1, l_2	link lengths	1 m, 2 m
I_1, I_2	inertia about centre of mass	0.083, 0.33 kg·m²
τmax	motor torque limit	20 N·m

Phase 1 — Modelling (week 1)

Find every equilibrium of the system. Linearize the dynamics around each one, and give the state-space model and the transfer functions from τ to each joint angle.

NOTE: Full Non-Linear Model will be given. Understanding of systems and equilibrium is tested here rather than your physics knowledge.

Submit by 25th October: a report of at most 2.5 pages. Correct any errors before relying on the model in later phases.

Phase 2 — Stability analysis (week 2)

Determine the stability of every equilibrium using both transfer functions and state-space analysis. Support each conclusion with simulation. At the upright equilibrium, determine whether feedback from a single measured joint angle can stabilize the system, and **explain why**.

Phase 3 — Control around equilibria (weeks 3–4)

Design controllers that stabilize the acrobot at the upright equilibrium and at least one other equilibrium. Justify your design choices. Characterize the limits of each controller: how far from the equilibrium it still works, with τ_{max} = 20 N·m.

Phase 4 — Swing-up (week 5)

Starting from hanging straight down, bring the acrobot to the upright equilibrium and hold it there. Whether your attempt works or not, explain with evidence why it behaved as it did: this analysis carries most of the marks for this phase.

Evaluation:

- Deliverables: A technical report (single-column, double spaced) and the associated Matlab code.
- Viva: An oral examination will be conducted, which will be based on your report more than the syllabus, so make sure you know what you are writing.

Final Notes:

- The Phase 1 grade is the one given in week 1.
- Technical analysis is prioritized over the controller performance.
- A working swing-up is not needed for full marks as long as you explain why in detail.
- Every claim in the report must be backed by a derivation, a number or a plot.
- A document will be released before every phase to help you with the project.